

# Analyzing ecommerce data

CASE STUDY: ECOMMERCE ANALYSIS IN POWER BI



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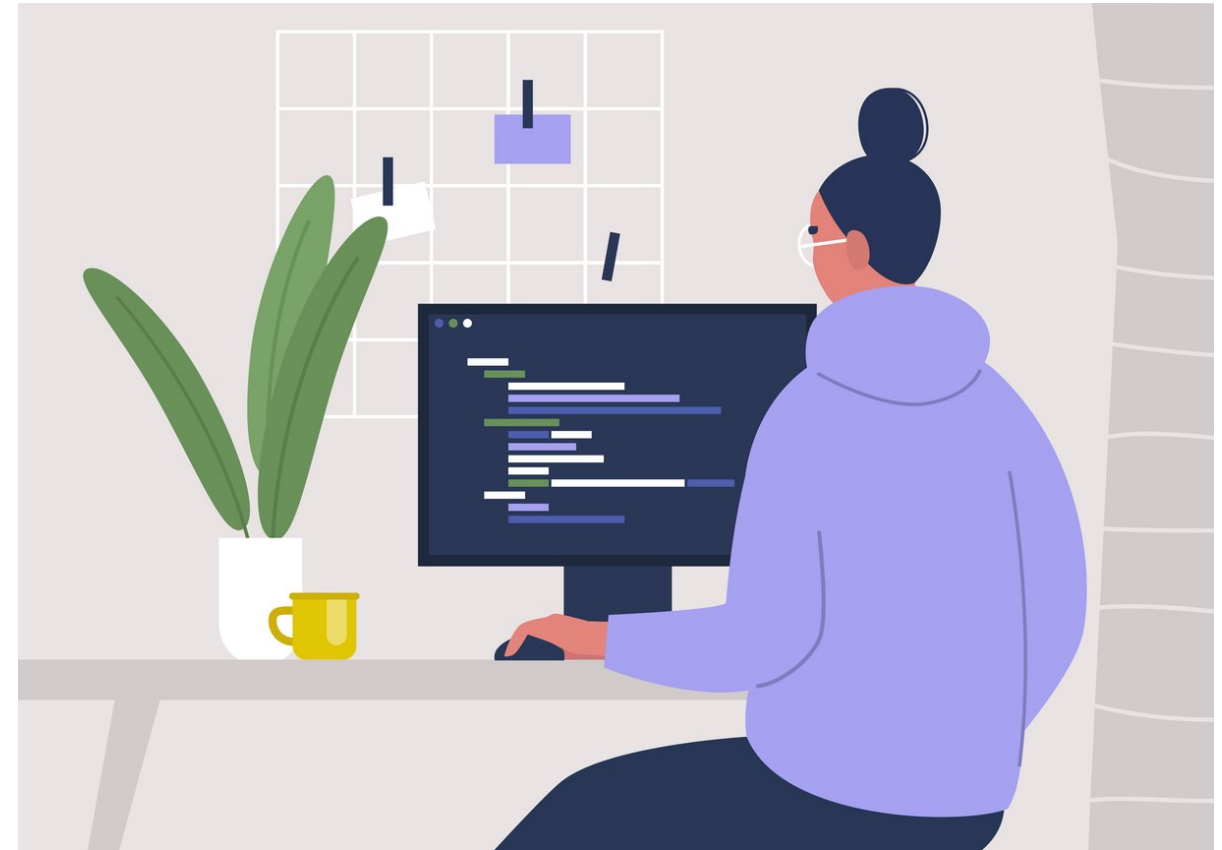
Content Developer, DataCamp

# What is a case study?

- Apply your skills
- No new concepts
- Solve a real-world problem

## Prerequisite courses:

- DAX Functions in Power BI
- Data Visualization in Power BI

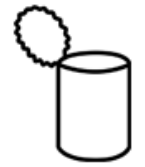


# Case study goals

- Fictitious data from online pet supply store
  - Whiskique
- Hired as a new analyst to provide insights
- Analyze an ecommerce dataset
- Leverage Power BI to create an engaging dashboard-style pages

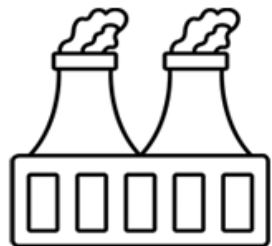


# Ecommerce operations and terminologies



Can pet food (4)

Illustrative total costs for canned pet food through various stages in the supply chain.



Factory

\$4

COGS  
(cost of goods sold)

+



Freight

\$1.5

Landed  
Cost (freight)

+



Warehouse

\$1.5

Fulfillment Fees

+



Shipping

\$5

Shipping  
Cost

+



Profit/loss

8

Profit  
(Loss)

=



Price

\$20

Retail  
Price

# Whiskique's business goals

## Primary Business Goals

- Serve as many customers as possible and increase sales
- Reduce operating costs



# Increase sales

- Cross-sell - promote a relevant product at the point of purchase
- Upsell - promote a higher priced alternative or a higher quantity of a product



# Shipping cost reduction strategies



- Shipment consolidation
- Package dimensions or weight
- Ship higher quantity

# Market basket analysis

- Find products frequently bought together
- Determine correlation coefficient between purchased items
  - Implemented as a correlation matrix
- Out of scope for this course
- Can identify frequency products are purchased with together



# The data source

| Sales table      |                                                                        |
|------------------|------------------------------------------------------------------------|
| Transaction Date | Date of purchase                                                       |
| Customer ID      | Customer identifier                                                    |
| Description      | Product description                                                    |
| Stock Code       | Product code                                                           |
| Invoice No       | An invoice contains multiple products and represents a single checkout |
| Quantity         | Quantity of a product purchased                                        |
| Sales            | Total amount of a product in a single checkout                         |
| Unit Price       | Unit price of a product                                                |

| State mapping |                                                |
|---------------|------------------------------------------------|
| Order State   | State code, description and all its variations |
| State         | a standarized state code                       |
| Region        | region name                                    |

| Product table      |                                                  |
|--------------------|--------------------------------------------------|
| Stock Code         | Product code                                     |
| Weight             | Weight of a single unit                          |
| Landed Cost        | Manufacturer cost + freight                      |
| Shipping_Cost_1000 | Average cost of shipping 1000 miles to customers |
| Description        | Most recent product description                  |
| Category           | Product category                                 |

| Customer table |                                |
|----------------|--------------------------------|
| Customer ID    | Customer unique identifier     |
| Order City     | City                           |
| Order Postal   | Postal code                    |
| Order State    | State                          |
| Latitude       | Latitude of customer location  |
| Longitude      | Longitude of customer location |

# Let's practice!

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